Proof by Induction: Sequences

Questions 1–13

Question 1. Show that

a) $3\left(3^{k-1}+\frac{1}{2}\right)-1=3^k+\frac{1}{2}$

b) $3(2^k+3^k)-2^k=2^{k+1}+3^{k+1}$

c) $\dfrac{\frac{3}{3k-1}}{1+\frac{3}{3k-1}}=\dfrac{3}{3(k+1)-1}$

d) $\frac{1}{2}(k^2-5k+12)+k-2=\frac{1}{2}((k+1)^2-5(k+1)+12)$

Question 2. Given that $u_1=8$ and $u_{n+1}=u_n-3$, prove that for all integers $n\geq 1$, $u_n=11-3n$.

Question 3. Given that $u_1 = 2$ and $u_{n+1} = 2u_n - 1$, prove by mathematical induction for $n \in \mathbb{N}$ that $u_n = 2^{n-1} + 1$.

Question 4. Given that $u_1 = 8$ and $u_{n+1} = 4u_n + 3$, prove that for all integers $n \geq 1$, $u_n = 9 \times 4^{n-1} - 1$.

Question 5. Given that $u_1 = 4$ and $u_{n+1} = 3u_n - 1$, prove by mathematical induction for all integers $n \geq 1$ that $u_n = \frac{1}{6}(7 \times 3^n + 3)$.

Question 6. Given that $u_1 = 2$ and $u_{n+1} = 8u_n + 5$, prove that for all integers $n \geq 1$, $u_n = \frac{19 \times 8^n - 40}{56}$.

Question 7. Given that $u_1 = 2$ and $u_{n+1} = u_n + 3n + 5$, prove by mathematical induction for $n \in \mathbb{N}$ that $u_n = \frac{1}{2}(3n^2 + 7n - 6)$.

Question 8. Given that $u_1 = 1$ and $u_{n+1} = \dfrac{u_n}{1 + u_n}$, prove that for all integers $n \geq 1$, $u_n = \frac{1}{n}$.

Question 9. Given that $u_1 = 2$ and $u_{n+1} = \dfrac{2u_n + 1}{u_n + 2}$, prove that for all integers $n \geq 1$, $u_n = \frac{3^n+1}{3^n-1}$.

Question 10. A sequence $z_1, z_2, z_3, \ldots, z_n$ of complex numbers is defined recursively as follows: $z_1 = 1 + i$ and $z_{n+1} = (1 + i)z_n + (1 - i)^n$ for all integers $n \geq 1$.

a) Find z_2 and z_3 expressing your answers in the form $a + bi$ where a and b are real numbers.

b) Prove by mathematical induction that $z_n = \frac{1}{2}(1 + i)^n + \frac{i}{2}(1 - i)^n$ for all integers $n \geq 1$.

Question 11. A sequence $u_1, u_2, u_3, \ldots, u_n$ is defined by the recurrence relation $u_1 = 3$, $u_{n+1} = \dfrac{u_n}{2 + u_n}$.

a) Find u_2, u_3 and u_4.
b) Find the first 4 terms of $2^{n+1} - 3$.
c) Suggest an expression for u_n in terms of n.
d) Prove by mathematical induction that your expression for u_n is correct.

Question 12. Given that $u_1 = q$ and $u_{n+1} = u_n + a$ for constants q and a, prove by mathematical induction for all integers $n \geq 1$ that $u_n = q + (n-1)a$.

Question 13. Given that $u_1 = 5$, $u_2 = 13$ and $u_{n+2} = 3u_{n+1} - 2u_n$,

a) Write down the values of u_3 and u_4.

b) Prove that for all integers $n \geq 1$, $u_n = -3 + 2^{n+2}$.